17.01.2024

EE2003 and CSE 3017 DIGITAL DESIGN FINAL EXAM

Name: Solution

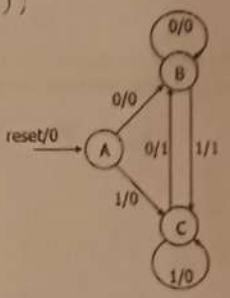
DURATION: 90 MINUTES
CLOSED BOOK, CLOSED NOTES EXAM. NO CALCULATORS OR CHEAT SHEETS ALLOWED.
PLEASE PROVIDE YOUR ANSWERS IN DESIGNATED PLACES.

$q_1\ q_0$
A: 00
B: 01
C: 10

Q1) For the finite state machine below,

(A) **(6 points)** Write Verilog description using case statements.

(B) **(6 points)** Fill the characteristic table.



reset	x input	$q_1\ q_0$ [n] current state	$q_1\ q_0$ [n+1] next state	y output
1	–	– –	– –	–
0	0	0 0	0 1	0
0	1	0 0	1 0	0
0	0	0 1	0 1	0
0	1	0 1	1 0	1
0	0	1 0	0 1	1
0	1	1 0	1 0	0

```
module final(x,clk,reset,y);
input x,clk,reset;
output reg y;
reg [1:0] state = 2'b00;
parameter A=2'b00,B=2'b01,C=2'b10;
always @(posedge clk)
begin
if (reset==1) state <= A;
else
begin
if (x==0)
case (state)
A: state <= B;
B: state <= B;
C: state <= B;
endcase
if (x==1)
case (state)
A: state <= C;
B: state <= C;
C: state <= C;
endcase
if ((x==0) || (state==C)) y<=1;
if ((x==1) || (state==B)) y<=1;
else y<=0;
end
end
endmodule
```

(C) **(6 points)** Write state equations.

$$q_1[n+1] = x\,\overline{q_1[n]} + x\,\overline{q_0[n]}$$

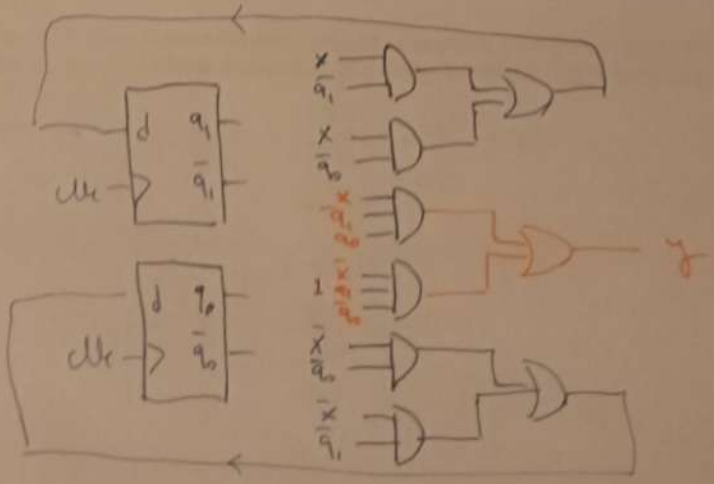
$$q_0[n+1] = \bar{x}\,\overline{q_1[n]} + \bar{x}\,\overline{q_0[n]}$$

$$y = x\,\overline{q_1[n]}\,q_0[n] + \bar{x}\,q_1[n]\,\overline{q_0[n]}$$

(D) **(3 points)** What does this circuit do? Explain.

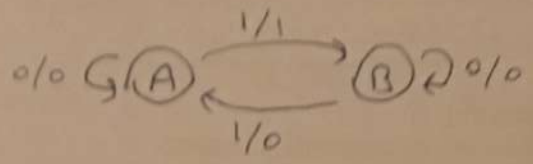
This circuit detects pattern BC or CB, in other words: 0110 or 1001

(E) **(6 points)** Implement the digital circuit using D flip-flops.



Q2) Design a circuit to detect occurrence of "01" in a stream of the serial data. Assume that input is x output is y.

(A) **(7 points)** Draw state diagram. A: 0 B: 1

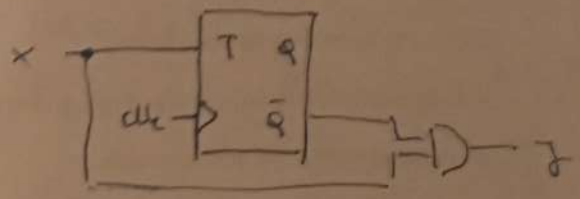


x	q[n]	q[n+1]	y
0	0	0	0
0	1	1	0
1	0	1	1
1	1	0	0

(B) **(7 points)** Implement the detector using T flip-flop.

$$y = x\,\bar{q}[n]$$

$$q[n+1] = x\,\bar{q}[n] + \bar{x}\,q[n] = x \oplus q[n]$$



Q3) For the combinational circuit below,

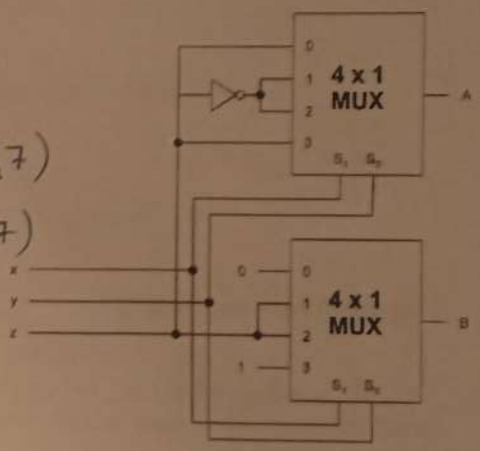
(A) **(5 points)** Determine the truth table for the below circuit.
(B) **(8 points)** Find minimized output functions A and B in SoP form using K-maps.

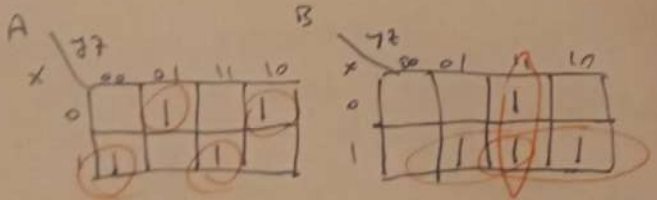
$S_1 = x \quad S_0 = y$

x	y	z	A	B
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$A = \sum(1,2,4,7)$

$B = \sum(3,5,6,7)$





$A = x\bar{y}\bar{z} + \bar{x}\bar{y}z + xyz + \bar{x}y\bar{z}$

$B = xy + yz + xz$

(C) **(4 points)** The circuit has the functionality of a commonly used arithmetic component. What does the circuit do and what are other names for A and B?

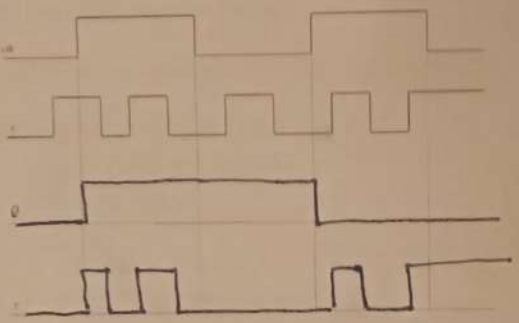
This circuit implements a full adder.
A is the sum
B is the carry out.

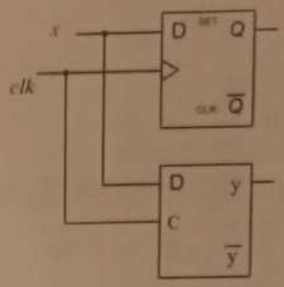
Q6) Answer following questions.

(A) **(4 points)** Given a synchronous sequential circuit with 17 states, the minimum number of flip-flops required to implement the circuit is 5 flip flops and the number of unused states is 15 states.

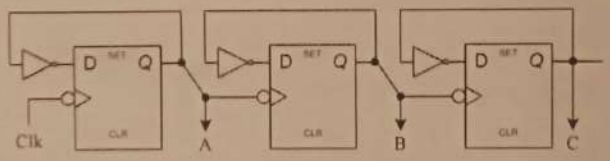
(B) **(4 points)** For a 3-bit synchronous binary counter (outputs Q2, Q1 and Q0), with input clock frequency of 32 MHz, the frequency of Q0 is 16 MHz and the frequency of Q2 is 4 MHz.

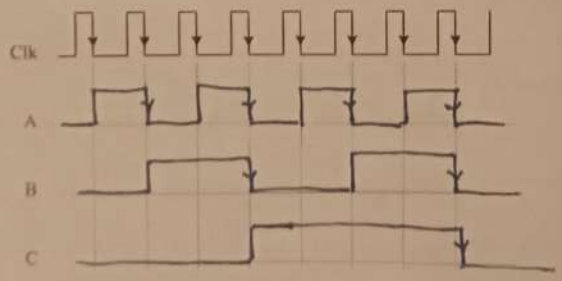
(C) **(4 points)** For the circuit constructed with a D flip-flop and a D-latch, sketch the output waveforms at Q and y given the shown input waveforms of the clock signal clk and the input signal x. (Note: Neglect propagation delays.)

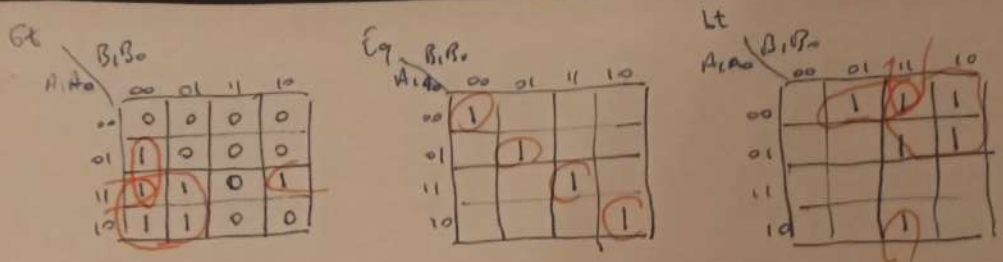




(D) **(6 points)** Draw the waveforms of signals A, B, C in response to the shown Clk signal assuming initial ABC value of 000. (Note: Negative clock input means falling edge triggering)







Q4) Design a magnitude comparator circuit for 2-bit binary numbers $A=A_1A_0$ and $B=B_1B_0$. The outputs are Gt, Eq, and Lt, where Gt is 1 if A>B, Eq is 1 if A=B, and Lt is 1 if A<B.

(A) **(7 points)** Form the truth table.

(B) **(7 points)** Determine the three minimized output functions of the comparator in SoP form using K-maps.

A_1	A_0	B_1	B_0	Gt	Eq	Lt
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

$Gt = \sum(4,8,9,12,13,14)$

$Eq = \sum(0,5,10,15)$

$Lt = \sum(1,2,3,6,7,11)$

$Gt = A_0\bar{B}_1\bar{B}_0 + A_1A_0\bar{B}_0 + A_1\bar{B}_1$

$Eq = \bar{A}_1\bar{A}_0\bar{B}_1\bar{B}_0 + \bar{A}_1A_0\bar{B}_1B_0 + A_1A_0B_1B_0 + A_1\bar{A}_0B_1\bar{B}_0$

$Lt = \bar{A}_1\bar{A}_0B_0 + \bar{A}_0B_1B_0 + \bar{A}_1B_1$

Q5) From file "ROM_entries_hex.txt" file, data is needed to be read. Location of the ".txt" file is given as,

```
4
5  initial $readmemh("C:/users/ROM_entries_hex.txt", ROM);
6
```

(A) **(5 points)** Compose a Verilog description for utilizing that ROM.

```
module my_ROM (data, address);
  input [2:0] address;
  output [15:0] data;
  reg [15:0] Rom [7:0];
  assign data = Rom [address];
initial $readmemh ("C:/users/ROM_entries_hex.txt", Rom);
endmodule
```

(B) **(10 points)** The contents of the memory is given below, write a testbench file that reads and displays data "a1fd" and "a3f9" from appropriate address locations.

```
a0ff
a0fe
a1fd
a1fc
a2fb
a2fa
a3f9
a3f8
```

```
module my_Rom_tb;
  wire [15:0] data1;
  reg [2:0] addr1;
  my_Rom UUT (.data(data1), .address(addr1));
  initial begin
  addr1 = 3'b010;
  #100 addr1 = 3'b110;
  end
endmodule
```